The Human Eye and the Colourful World

The big idea

The eye is a natural camera with a lens that changes its focal length. In this chapter we see how the eye works, why some people need spectacles, and why the sky is blue and rainbows form.

Structure of the eye

Part	Role
Cornea	Transparent front; most refraction happens here
Iris	Coloured muscle that controls pupil size
Pupil	Opening that lets light in
Eye lens	Fine-tunes focus to form an image on the retina
Ciliary muscles	Change the lens's curvature
Retina	Light-sensitive screen; cells send signals via the optic nerve

POWER OF ACCOMMODATION

The ciliary muscles make the lens **thinner** for distant objects and **thicker** for near ones. The **near point** of a normal eye is about **25 cm**; the far point is **infinity**.

Defects of vision

Defect	Can't see clearly	Cause	Correction
Myopia (short-sightedness)	Distant objects	Image forms **in front of** retina (eyeball too long or lens too curved)	**Concave** lens
Hypermetropia (long-sightedness)	Nearby objects	Image forms **behind** retina (eyeball too short or focal length too long)	**Convex** lens
Presbyopia	Near objects, in old age	Ciliary muscles weaken	Often **bifocal** lenses

Cataract (cloudy lens) is treated by surgery.

Refraction through a prism

A glass prism bends light twice, so the emergent ray is deviated from its original direction by the **angle of deviation**.

DISPERSION

White light splits into its seven colours, **VIBGYOR** (violet, indigo, blue, green, yellow, orange, red), because each colour bends by a different amount. **Violet bends the most, red the least.** Isaac Newton showed that a second, inverted prism recombines them into white light.

A **rainbow** forms when sunlight is dispersed, refracted and internally reflected by tiny raindrops. You see it with the Sun **behind** you.

Atmospheric refraction: hot air above a fire makes objects behind it seem to flicker; stars **twinkle** because starlight is refracted by layers of the atmosphere; we see the Sun about **2 minutes before** actual sunrise and after actual sunset. Planets don't twinkle because they are closer and act as extended sources.

Scattering of light

Tiny particles scatter light. The **Tyndall effect** is the scattering of light by particles in a colloid, which makes the path of a beam visible.

WHY IS THE SKY BLUE?

Air molecules are much smaller than the wavelength of visible light and scatter **shorter wavelengths (blue)** much more than red. This scattered blue light reaches us from all directions.

At **sunrise and sunset**, sunlight travels through more air, most blue is scattered away, and mostly **red** reaches us. Danger signals are red because red scatters least and travels farthest in fog.

Quick recap

- The eye lens changes shape (accommodation); near point 25 cm.
- Myopia → concave lens; hypermetropia → convex lens.
- Prisms disperse white light into VIBGYOR; violet bends most.
- Scattering explains the blue sky and the red sunset.

Practice questions

1. Myopia is corrected using a
   (a) concave lens (b) convex lens (c) bifocal lens only (d) cylindrical lens
2. The near point of a normal eye is about
   (a) 10 cm (b) 25 cm (c) 1 m (d) infinity
3. In a spectrum of white light, the colour that bends the most is
   (a) red (b) green (c) violet (d) yellow
4. Stars twinkle because of
   (a) dispersion (b) atmospheric refraction (c) reflection (d) scattering
5. The sky appears blue because
   (a) the ocean reflects blue light (b) air molecules scatter blue light more than red light (c) the Sun emits only blue light (d) of dispersion by clouds
6. The part of the eye that adjusts the size of the pupil is the
   (a) cornea (b) iris (c) retina (d) ciliary muscle

Answers: 1. a 2. b 3. c 4. b 5. b 6. b